

CTL 密钥封装算法 x86 架构实现

自评估测试报告

算法名称：CTL

算法类别：密钥封装

细分功能：密钥生成/封装/解封装

实现版本：优化实现版

安全级别：128bit/256bit/512bit

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1. 概述

本报告针对 CTL 密钥封装算法在 x86 架构下的参考实现版和优化实现版开展自评估测试。测试内容包括功能测试、性能与资源消耗测试、传输与存储开销测试。

功能测试用于验证 CTL 算法密钥生成、封装、解封装等核心部件是否完整，验证封装端生成的共享秘密值与解封装端恢复的共享秘密值是否一致。测试过程中采用固定随机种子或固定测试向量，保证测试结果可复现。

性能与资源消耗测试用于统计密钥生成、封装、解封装三个核心操作的平均耗时、平均 CPU 周期数、吞吐量，以及算法加载后静态内存占用和运行过程峰值内存占用。测试次数应不少于 100 次，实际测试次数根据测试命令填写。

传输与存储开销测试用于统计 CTL 密钥封装算法涉及的数据尺寸，包括公钥、私钥、密文和共享秘密值等。

2. 测试环境

测试时间为：2026/6/24-2026/6/25

2.1. 软硬件配置

序号	名称	规格和标识	用途	数量
1	CTL 参考实现版程序代码	版本：__1.0__；参数集：__257-512/769-1024/3329-2048__	被测试程序代码	1
2	CTL 优化实现版程序代码	版本：__1.0__；参数集：__257-512/769-1024/3329-2048__	被测试程序代码	1
3	测试终端 PC	CPU：__Intel® Core™ i9-14900HX__； 主频：__2.2GHz__； 内存：__32 GB__； 系统：__Windows11__	编译与运行测试程序	1
4	MinGW 编译环境	GCC 版本：__15.2.0__；目标架构：x86-64	编译 C 语言测试程序	1

2.2. 测试工具

序号	工具名称	工具类型
1	MinGW / MinGW-w64	C 语言编译工具
4	benchmark_selfeval_ctl.c	性能、资源消耗、传输与存储开销测试程序
5	test_kem_correctness_ctl.c	功能正确性测试程序

3. 优化实现测试

本章针对 CTL 密钥封装算法优化实现版开展测试。

优化实现版使用如下编译命令：

```
gcc -O3 -march=x86-64 -mavx2 -mtune=native -flto -fomit-frame-pointer -std=c99 -Wall -Wextra -Wpedantic
test_kem_correctness_ctl.c <CTL_SOURCE_FILES> -o test_kem_opt.exe

gcc -O3 -march=x86-64 -mavx2 -mtune=native -flto -fomit-frame-pointer -std=c99 -Wall -Wextra -Wpedantic
benchmark_selfeval_ctl.c <CTL_SOURCE_FILES> -o benchmark_opt.exe -lpsapi
```

3.1. 功能测试

测试编号	OPT-1
测试名称	CTL-257-512 优化实现版密钥封装功能正确性测试
测试目的	测试 CTL-257-512 优化实现版是否支持密钥生成、封装和解封装功能，并验证封装共享密钥与解封装共享密钥是否一致。
预置条件	1.已在测试终端 PC 上部署 CTL 优化实现版程序代码； 2.test_kem_correctness_ctl.c 能够正常调用 CTL 的密钥生成、封装、解封装接口； 3.测试程序可正常编译并运行。
测试步骤	1.进入 CTL-257-512y 优化实现版程序代码目录； 2.输入 make all 生成 test_kem_correctness_ctl.exe 文件； 3.运行 test_kem_correctness_ctl.exe； 4.比较封装共享密钥与解封装共享密钥是否完全一致； 5.重复执行 3 组测试，并生成 ctl_kem_kat_vectors.txt 结果文件。
预期结果	1.test_kem_correctness_ctl.exe 程序能够正常运行； 2.终端显示 “All CTL KEM correctness tests passed.” 3.正确生成测试向量文件 ctl_kem_kat_vectors.txt。 4.测试向量文件中的 ss_enc 和 ss_dec 相等；
实测结果	程序正常运行且显示 “All CTL KEM correctness tests passed.”， ctl_kem_kat_vectors.txt 文件中的三组 ss_enc 和 ss_dec 都相等。

测试向量：algorithm = CTL-257-512

```
seed_len = 55

initial_seed =
ADBE8B8491E2FFC8C5D6233C091A17607D4E5B54A1B28F9895E6F3CCD92A27300D1E6B6471425F
A8A5B6839CE9FAF7C0DD2E3B3401126F

case = 1

pk_len = 521

sk_len = 2953

ct_len = 473

ss_len = 16
```

pk =

99B4BD408EE8A95011ACEA41FEE592AF5879CD3D3EAD420122FB50AC7B94D5F15091E56DA6D
57C7A10E2766DF8144E94A2D67DE507A6139B0DA5DC0B72A106A1FA49413C8612D88938AC9316
DC8A0EFB45464AD04FDA893F76B94333FDA9976DFBFD39CDC7CDB2BC071FD0F98D5551D5D35
A4CF0EFF3C04BF4EAA49013E3A3586A562C012D6CEBB7DCB48F1B310F471365D884FD22C9217F
36615FA02CD2082B0C33D7CA23014D918331530BE5E0D00598F7AF0B93E55A9151146B3D80889682
1903579022FF90557DBA44F95E401DD029E325F672D50CC90F0FEE91799F3F49E44FCEC1330B8EFC
FA0BB0A10775A7410758235182FA57D2E9455E40C6C393502FEDA2F92745D1FD0EB1A39786C9FB
DA9B9202F4E3E791EA7DC7D953AF696588FE47907C8A1A908F5D2EAE969A9421029A9F7A664DB
8B81D059289D53F98E164C65AB0C10641641211590A98B32B56EBD084F17B58F3B823B62EA25FC1
FAAF00D6BABF8F807BFAF6062E8CFA2A79872931B5BE46164F97E0DAD625C100521488E6331E93
F9024946CB7E7D2ED3C6B7CFBF164A0D36E1521DD2EE461B9DE28AF9019A85D33DE8C72BC75E
97E47E97DC02A7F1D94B3C3D54B994B59D2F48C30B6DDCFF0253E781E573CD09BFC1D20076C45
87A4F85009894AD762C19D7EE5FD0356B546F654705DEAB6426B0916A05057E176F2A00C5B2D349
00920AD9FE9510F147A389E5BCD6BA907

sk =

19AE5D933F84538C0270359F7DED4695A38B68A9E050084EFBA8BDCDD685B942EE3FE05922EE90
1FE0E9E956242B81B7F61C15C1111D711EAEB4076508CEC087DD0000F011FF00F0011010000F00001
F1000000000100010F0F10F11F000101F0000F00F011FF100000000101FFF01F110FF000F00F000F100
0F101F00000001010F00011F00000F010101110F00F0000F000000110000000000000F0F0100FF0000001
F00F000000001001000100110F1000F00FF0F1F0F00F0010000010FF00FF000F0000F00F001F0000FF001
0011010000000FF01100F0F0F1F00FF00010100000101011F000100FF010F0F00001F00F00FF0F0000F00
00101100FF0001000F10001F010000F001F0001F0F0000F00F001F1F000F001000F00F110F00F000000F0
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000F0F0000001F000100F10100000000000100F10100000000000F01000F0000000FF0FF010000000001
00F000FFF100000F00FF10001000010000F10001F01001010F0000FF110000F000000100100101100010
00110F010000000111000011F0010F0001F1100F00F0000100001000F00000010F1000F000001000001100
000F10FF011F0F0101100000101100F0F000000F010011100F00010000100010010101FF1010F001000000
000F000000000F0F0010111101010F000100010000F0F0010000000F10000F00000FF000F00100F01F00
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5FC1EFC079FBB2011880BFF7AFFCEFB042F44140F84EBFF03001E44083100FC6EFFEFEBFAF00EC3
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F40E7CF3F07F13D041FC3F80F3FFC1FC6F7F18AFBF03D03CF80FC1F3D07BFFE07E17A04300307BF
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FB7F782BBCC6D4E5D28E0422935ECEA1941B347259A2D8B98B263F515ADB3F4E601340333E26240
7157EE88DA7B97EE127E64E2F4C35D894F2107D14C1E93BA9AE7C283F500D59D4B72D1D6516453
FDF03224377464F95C3D905374F1B9F32AC7F4F7DC5C8895381C272DBC90A9896BAEE611A5EEC5
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9B545C6939C0677AD6453527DA1C0BB8E51C13DD7BE850CB1FE3C55D39300EC10EC8E4E23EA2B
90B8BAC11A16F2B9E1E4E4DDE8477FF3D384C535ECFC4B7DDAA077CF43AFD9B2D2D50EB40F6
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985D589F0E03D7347953C1DEE42A7698FA40A259F102BB46EBC14C3AF080E78212D9C5E236C6AE
74D6EB961AC1F2F50F3B8CE74EE0DC098877C095DB98D7C2EEF936BB641A13CD375B975E8650E
E10C6178BBB594528CD07FE2414F50C28D3FA846D85F727A3422F8A593648C711902929D0F9B4355
6C61F2FE5D19D98FB2C035DE1207B868C167B79A3A2C8DBB442A5DD0F2E1D881EC644823F4ECC
261EFF4C447B5901FF7D99327B256010BD9E3A08355C4D1A16609B577A086669A300695960CF0645
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2AD5937CBE61C4760C93002AEFB87DCB3CD75B9A37021A865DE1DEF7B2FF8A418AE3FD019E6D
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275F1B4970ABA14B6EC75E9697FB461CAF12EE92ADB6AD023C31560F5AC457A792F5C77598F7B4
D9023080C689C484C8887AD8E1B9E191B6AB7E55FE441E88DFC64BA36C16D8EF91F27902A8F1EE
CE20E7259230D0D06F16D3023E7CBC8D897192AC1F2FDB8B0C53025F082324E99283C1901BF025B
0AB1DD3957A787A3A41426F763FC86598105816CC8F2FCE1DC6F4CD7547020F4E83FC6AC2906D2
1B38C81D33F4C31B628229BDF3C334312A541C7DBFD9EA72886E45C3FC3CE28CD84253641FBB02
B4BD408EE8A95011ACEA41FEE592AF5879CD3D3EAD420122FB50AC7B94D5F15091E56DA6D57
C7A10E2766DF8144E94A2D67DE507A6139B0DA5DC0B72A106A1FA49413C8612D88938AC9316DC
8A0EFB45464AD04FDA893F76B94333FDA9976DFBFD39CDC7CDB2BC071FD0F98D5551D5D35A4
CF0EFF3C04BF4EAA49013E3A3586A562C012D6CEBB7DCB48F1B310F471365D884FD22C9217F366
15FA02CD2082B0C33D7CA23014D918331530BE5E0D00598F7AF0B93E55A9151146B3D80889682190
3579022FF90557DBA44F95E401DD029E325F672D50CC90F0FEE91799F3F49E44FCEC1330B8EFCFA
0BB0A10775A7410758235182FA57D2E9455E40C6C393502FEDA2F92745D1FD0EB1A39786C9FBDA
9B9202F4E3E791EA7DC7D953AF696588FE47907C8A1A908F5D2EAE969A9421029A9F7A664DB8B8
1D059289D53F98E164C65AB0C10641641211590A98B32B56EBD084F17B58F3B823B62EA25FC1FAA
F00D6BABF8F807BFAF6062E8CFA2A79872931B5BE46164F97E0DAD625C100521488E6331E93F902
4946CB7E7D2ED3C6B7CFBF164A0D36E1521DD2EE461B9DE28AF9019A85D33DE8C72BC75E97E4
7E97DC02A7F1D94B3C3D54B994B59D2F48C30B6DDCFF0253E781E573CD09BFC1D20076C4587A4
F85009894AD762C19D7EE5FD0356B546F654705DEAB6426B0916A05057E176F2A00C5B2D34900920
AD9FE9510F147A389E5BCD6BA907

ct =

D98D3DAAFB6D25886B47F191AA1E96323CB0B48B185E1B7F0DAB551ADE7C6A0B7DF36741428C
BC7DE0FC30FEF88B3F62F4C11C82E28A7BA66E94BC4B099E72015F2CE8A95EC2A54371FA6535F1
696A7D385D9373B2705133E05CE35E3D8D350E5B06CC20CC451368F31D1C6E8D29B7F63C7F6CCA
2A19397E594B6FC51388B3A575C29A25BEEE271ADCE66477ABC252B7DDB9ADC3151663D8499EB
3FE32FBBE8EBB722E47DC16F542D139B939B635C1A77B9245E84EB4FA7DE57494A9EF6674707763
4CBFE531C6D231C23A292E39B4A3BCF5DDDEAF27463F2AA1BAC9B643A840E93FCC54F4323F33E
B780A8AB5309112A27C953241AA31A78325D41204F0AEA2F7B8FB2C2EA3111831DF8052347E3C9
C32C8FB6A06B1F663D2781299AD9166F023900EAD92DEF85D3F05F201E9C4688C934E00C22F4062
9D59445A4CFE6F739D72B8AE7851473BB20B7DE12E167D0F9EC51FEF9F9467E21C5B0FFA09EF88
8F319CC8311745E685099A7CEC230B73F937E496B2D60B39F7FA2B419678B70A02114A91843DE800
F7AA58F6FFFA992AB3F847CC0A1E4B54DBCA405E65BBF0D577D01AD73F84D3E801A99157AC2A
3A3AE1E78A39284A9A0736242014B56AA4147D930F8C2826D080B9CE114A873906B90B1AE9E9876
E054B8D0C284

ss_enc = 460FE91C8D752E7BA74838C7B2B891DD

ss_dec = 460FE91C8D752E7BA74838C7B2B891DD

result = PASS

case = 2

pk_len = 521

sk_len = 2953

ct_len = 473

ss_len = 16

pk =

991581121AADB24C53C4E3C3ED26E30087DDA5E64C6C24CE94963F7C4BE3EBEA5C89F63B141F3
B9B1D37FB499539AD64C53B5502F1680A15631F6C98F730431373C9A54EFBA63DAB090E9725C196
1EE1CD771F4FB93A9E5786D8F1FA74BF773C4C4E48384E6CAE5392417570B9B85E49E1800BC4469
7A73EEF59C550B34D8509FD3E3B878A74A3BEA2E681C900DB223F2BDE9CF48A56A945729486E7E
A53567320E28719AB60AB6DC218C85822BD0E5814AB3828656A180B93F22B6BEE3851234EE91440
7E4B59A6E7D59A604C8CACD5233B5EE7D5DEBCAF0FB8949C917DA888C0A366767058E64022C0F
6DBE36AA3DE5E0EC9C9891C11E86319FB71090EF54E588B3A30C14626FAB7888FD04440C94A178
B7FA5590D86C40CF4F9A8FC5D97E7BEAF4898385C05FE44C0560D7137DADEAE8BEA2A003F406
AE1B7E66F847080A86F56ECE1142826DBCC33D392F46495CB9B6DA18C5A16E38DE256C3CA1F20
AD186A4207A91B5EC6BBC546C963B9D645CA14C5A8255935277F87FC4993F6AEACCEE1550915C
4328A34F2D50E6A758035CC20718A915A679B31460BA273B3F64B19DBC8372A0999949AD412C290
E3205752F8AC15AAECF02D06F776FDD508418B8DA406229E53301F1170D1D026F8562B13E3B6C10
CDDBC2266F66AF635742D3E3568ABF55F0B437B732B81FE89A9D4A126DEC5EA5CEE266EF9DA
60851CCBD3E522E3429600E8D552AD2

sk =

19C10DA35D778BACBD25A9A47315A4F312EFD4EC044C315DC4CA00F59E297A9F8D1A79ACA146
51B4FC7482B26D2F7125FCBEED20025EE7947A5AE5FE04F9B6681EFFFF000F0F0000000001F1FF1
00F00F0000F100000000F00000010010000000000F02F0F0F10010000FF0000F00F1000F1000F0100F
0010F00000010000000000F001010001F0F0100F000F01E111000010F00100000F000F010FF1000000F
001000FF0110100011000F01F0001F0F00000F000100100000F01F010F0F00000000FF10000F00000000
01000010F000F10000F0000001F000001010F10F1000F00FFF00100000F0F000F000F0F00F0F0000F00F0
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result = PASS

case = 3

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sk_len = 2953

ct_len = 473

ss_len = 16

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6FE104B2BFB2E3B171C7109FE7245A34E73D301BC1F6A07E129A109FD182681A5780E7C8A3AB6D
039ECFE8753AF3FC83EE0C6D52E79BD47B6F1522384B42E49ADA156F2EBC6762CB3B32201C79E4
9470D978268353A553162CA343B3FD01BF33E1EBC7141DCA9F19E2DD103220921CC8EBD905E58C
804C9F632DD3560BF78A21513DE403BDEDF4AA95EFE492C96AB37B739263ACE66BF0383915A036
D9F97E91D1F285C87688B0B1E793B8EC478517DFE074B3D456896C53392A846259B88C45BF71AF8
1200DA5853B796F9117AD03450DF117073203B1EBD19F7B67207808F6F71CCA8ABC211921C98F8F
96C6D1B29

ss_enc = 28A57455611066AE6E52E5E7A2604445

ss_dec = 28A57455611066AE6E52E5E7A2604445

result = PASS

测试编号	OPT-2
测试名称	CTL-769-1024 优化实现版密钥封装功能正确性测试
测试目的	测试 CTL-769-1024 优化实现版是否支持密钥生成、封装和解封装功能，并验证封装共享密钥与解封装共享密钥是否一致。
预置条件	1.已在测试终端 PC 上部署 CTL 优化实现版程序代码； 2.test_kem_correctness_ctl.c 能够正常调用 CTL 的密钥生成、封装、解封装接口； 3.测试程序可正常编译并运行。
测试步骤	1.进入 CTL-769-1024 优化实现版程序代码目录； 2.输入 make all 生成 test_kem_correctness_ctl.exe 文件； 3.运行 test_kem_correctness_ctl.exe； 4.比较封装共享密钥与解封装共享密钥是否完全一致； 5.重复执行 3 组测试，并生成 ctl_kem_kat_vectors.txt 结果文件。

预期结果	1.test_kem_correctness_ctl.exe 程序能够正常运行; 2.终端显示 “All CTL KEM correctness tests passed.” 3.正确生成测试向量文件 ctl_kem_kat_vectors.txt。 4.测试向量文件中的 ss_enc 和 ss_dec 相等;
实测结果	程序正常运行且显示 “All CTL KEM correctness tests passed.”, ctl_kem_kat_vectors.txt 文件中的三组 ss_enc 和 ss_dec 都相等。

测试向量: algorithm = CTL-769-1024

seed_len = 55

initial_seed =

ADBE8B8491E2FFC8C5D6233C091A17607D4E5B54A1B28F9895E6F3CCD92A27300D1E6B6471425F
A8A5B6839CE9FAF7C0DD2E3B3401126F

case = 1

pk_len = 1230

sk_len = 6030

ct_len = 1006

ss_len = 32

pk =

AAEDB9E409B79C235DEBA6776DAB251B5E7B60A973DDFAEC4C60ACC7529CA05E43356A1FB9
B066A2C9773E8E084FE2FCC05326C2C93DAA35B0A4145BC72AC251CB8BD997ED518CAEF173811
FED4E5BF320AF90670B5F8FDD7949E4C62EFF971F575B76D80582E3E29352DBE23AE1AA85205C1
6C849DBAF150FB8E831E018A503C24781F00F3EBF9D34F32C38DD71572794D152102F5AB2DFD3E
F74EBCB4511151C61A6ABE00263AE7EC0B717CF11D5F56F1A6F291E176ED9DB1E4653ECFF719F8
B8EEBE16419EA4DF190604FF5EBAFE4D056CEA3B82D1F82B915581559175E0658682906A377A30D
B0CD67F1708CC7C62C00BD9E8754E7A2433A94EDC2F94546916ECE5916E822FB478E941046C2F0
5B8D09FA1AFF6343B0FA9D472B28FF30383A70ADEBDC9DB655AABE051A81B7679AA875509571
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05486C2D07CC650DC876A91C31159D72E3152D3D25CA5D141A2E6D3B48435BA64738C0E606E883
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885133F9C7911DE1A7A8F6C5479689BD348116115FC7611DF8ABC70B0FF5409D41435A5783E0F292
177B55BAB65576997165C50636929CC24000EA99C99A619B4104F1D384CC839A834FD3CCABDAF3
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79DFAC50C108109CBBD0D4364833DC6AA422F100CFC195FDF9DFDBEBEDD297F3D0F9783B7AE
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BB4B820DF1A3AB8934BE17BF52F110CA57C7A6A9AD6DF8594F51608164BFD0601B07870B864FE8
B1B088D11031E51910D1E705027AEAF1FBEE109773709A3EE7298A0319DCB7720870C3C03755FFE
9A332F0D73943078EFD56367A98702B582BBD7A4BC44649C9C8CFC584E3A9A29E95E24A73445
75DD2421CA8834BA68027BAD42D2C26AB847E68CDC1B7E6D5EAB2D5501044685CFBC9D90948E
A3199FC10902617964B454F0BFA290CE439721374A59556BB01B1

sk =

2AE87E15F7E6CAB66514C99994C2F5BD92BC52A8554D2C081C0FDE814A57C2964B88F2D26D1A7
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result = PASS

case = 2

pk_len = 1230

sk_len = 6030

ct_len = 1006

ss_len = 32

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result = PASS

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ct_len = 1006

ss_len = 32

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ct =

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68905DBB91389A0D3BB2B333D14D9CFC3A2041B45B4CE500BED723F20E2

ss_enc = B11EF9E97DDC18BE412371D4279CB7626625E9B8459D5E25B0D625C289DDAAC

ss_dec = B11EF9E97DDC18BE412371D4279CB7626625E9B8459D5E25B0D625C289DDAAC

result = PASS

测试编号	OPT-3
测试名称	CTL-3329-2048 优化实现版密钥封装功能正确性测试
测试目的	测试 CTL-3329-2048 优化实现版是否支持密钥生成、封装和解封装功能，并验证封装共享密钥与解封装共享密钥是否一致。
预置条件	1.已在测试终端 PC 上部署 CTL 优化实现版程序代码； 2.test_kem_correctness_ctl.c 能够正常调用 CTL 的密钥生成、封装、解封装接口； 3.测试程序可正常编译并运行。
测试步骤	1.进入 CTL-3329-2048 优化实现版程序代码目录； 2.输入 make all 生成 test_kem_correctness_ctl.exe 文件； 3.运行 test_kem_correctness_ctl.exe； 4.比较封装共享密钥与解封装共享密钥是否完全一致； 5.重复执行 3 组测试，并生成 ctl_kem_kat_vectors.txt 结果文件。
预期结果	1.test_kem_correctness_ctl.exe 程序能够正常运行； 2.终端显示 “All CTL KEM correctness tests passed.” 3.正确生成测试向量文件 ctl_kem_kat_vectors.txt。 4.测试向量文件中的 ss_enc 和 ss_dec 相等；
实测结果	程序正常运行且显示 “All CTL KEM correctness tests passed.”， ctl_kem_kat_vectors.txt 文件中的三组 ss_enc 和 ss_dec 都相等。

测试向量：algorithm = CTL-3329-2048

seed_len = 55

initial_seed =

ADBE8B8491E2FFC8C5D6233C091A17607D4E5B54A1B28F9895E6F3CCD92A27300D1E6B6471425F
A8A5B6839CE9FAF7C0DD2E3B3401126F

case = 1

pk_len = 3009

sk_len = 15617

ct_len = 2353

ss_len = 48

pk =

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ss_enc =

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ss_dec =

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result = PASS

case = 2

pk_len = 3009

sk_len = 15617

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ss_enc =

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ss_dec =

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result = PASS

case = 3

pk_len = 3009

sk_len = 15617

ct_len = 2353

ss_len = 48

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A6AE241166208948910522D47DE47C6BF17B332FDDA2734EB10B6DF702952D5C8360E645D3A83A
2D575BA30D0852A45DE3DD0D2A1DEB0590FD3782047AEAFB67B7813BFC2097693F0182DF70095
0DD311E86B440F81340EBA285F46A5AF7414925663FD2B08F20190244637A388663BE9062417A5584
3F4B251E4CBFB7536660B007463B623F3221A4F877DB16100EF8872CE512486EFA97987566E1CE0A
46953D33A24CC4941F03E166C0062DC2101C0862E171B32A65F15541D50211A40591F6921AD6E7A5
6221320E547124B57562F1209A49561A46F9A3FF5CB5336DA7B97D61EB17752D6D9D805B95E1C71
81C82B78E96D77F8571F5757BC86FC1CAC234BF0E5048F2E078D5C694226D6425A4B87B6FB389E0
E6AE0361D8BB282F1BECF9686244D0365435CCC2A89E788D942558432E06381A061F4B27A3FFC03
2BF64E03C3FBCF47091FBB25145B61BE9F88D4F557819105967E2FA66FE10575C43A7037991DDB97
489B52F495C00F4EB445CF4E2217F84991722D46A1D50F65A9CEB112901B1401E15CD9D238EEC81
A6C6D2C4B87F76D0A157CFD2D996418FA79227583522EE365BB42ED115524FAB34851DF611DE6B
0F42036BAE7593F44C392C424D506934F3D055630328BA2F425103892915A49413A1ACD74AA91FC8
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F43AD2A727EA53F1429C7AE542CA912F015C0B6B1DD1A626394611BBFCF0D2CD022A24475B75D
5B50DE22C1203CA35ED2657738E1D4D5EEBEA2317A943CC467D371297D36001D5797E599D63423
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27A3DD52FC1B2813899D315F29A4D5B1C769A7FC3F2B1EBBD834074AFC23251E5AEA43C75F58A
320B902399167045851704F1843B2DE12A199191B4FCF1083F87D1135AD9B8A3071613DF32FD4AAB
36772F7027C6C93A251A3B477D9D2716224BA45C72984A047E85C7F108A5995281928B162F05079E1
48694813EC765FCE92A52F52C972E9085AD7D432968AAACE23EC11DF545F139E0A1C2B4A335738C
815E3045B024130E2818B133CE3B60808BE0EDB987070E2C5C0625E454272FE1DC371CDDF931D41
5F0263817EB548FC55F9E4B0A235BE31395D986EE19529D1466E50C487A0709FA63C5C85707745276
204B5534F12165E560CC1813A10914995260FCB669758B3C57CA4B76CA3B77D710771C75F214C4F5
D7DFDB8C22B3C9DCC2233FA2120A381AEE02325B5FFF124EFAF09611B3859F0D9ED2D3FC9484D
A9095830D73B21EBDACF132D592D8CDA9406AE4FA7D3859EE5A654B00871E40

ss_enc =

0FA47F93CE8DAD32F2FE0D7E78D147DA29BEDDDD6F0BF2D4AE7AA192C3F014DEC7FA458B69
ADA01CDDC0338FDA62B9C6

ss_dec =

0FA47F93CE8DAD32F2FE0D7E78D147DA29BEDDDD6F0BF2D4AE7AA192C3F014DEC7FA458B69
ADA01CDDC0338FDA62B9C6

result = PASS

3.2. 性能与资源消耗测试

测试编号	OPT-4
测试名称	CTL-257-512 优化实现版性能与资源消耗测试
测试目的	测试 CTL-257-512 优化实现版密钥生成、封装、解封装三个核心操作的平均 CPU 周期数、吞吐量、静态内存占用和峰值内存占用。
预置条件	1.CTL 优化实现版程序代码运行正常； 2.benchmark_selfeval_ctl.c 能够正常调用 CTL 算法接口； 3.测试程序可正常编译并运行。
测试步骤	1.进入 CTL-257-512 优化实现版程序代码目录； 2.输入 make all 生成 benchmark_selfeval_ctl.exe； 3.运行 benchmark_selfeval_ctl.exe 程序分别统计密钥生成、封装、解封装的总耗时、平均耗时、平均 CPU 周期数和吞吐量； 5. 程序在算法运行前记录静态/基线内存占用，在多次执行过程中记录峰值内存占用,并最终生成 ctl_kem_benchmark_raw.csv， ctl_kem_benchmark_summary.csv 结果文件。
预期结果	程序能够正常运行，并输出算法的性能测试结果。包括密钥生成、封装、解封装操作的平均 CPU 周期数、吞吐量、静态/基线内存占用与峰值内存占用。
实测结果	程序能够正常运行，并输出性能测试结果。程序分别统计密钥生成、封装、解封装操作的平均 CPU 周期数、吞吐量静态/基线内存占用与峰值内存占用，测试次数为 100 次。 密钥生成平均 CPU 周期数为 7358433 cycles/op，吞吐量为 328 ops/s； 封装平均 CPU 周期数为 19383 cycles/op 吞吐量为 125407 ops/s； 解封装平均 CPU 周期数为 88852 cycles/op，吞吐量为 27250 ops/s。 静态/基线工作集内存平均值为 5559042 Bytes； 峰值内存平均值为 5562368.00 Byte；峰值内存平均值为 5717524.48 Byte；

测试编号	OPT-5
测试名称	CTL-769-1024 优化实现版性能与资源消耗测试
测试目的	测试 CTL-769-1024 优化实现版密钥生成、封装、解封装三个核心操作的平均 CPU 周期数、吞吐量、静态内存占用和峰值内存占用。
预置条件	1.CTL 优化实现版程序代码运行正常； 2.benchmark_selfeval_ctl.c 能够正常调用 CTL 算法接口； 3.测试程序可正常编译并运行。
测试步骤	1.进入 CTL-769-1024 优化实现版程序代码目录； 2.输入 make all 生成 benchmark_selfeval_ctl.exe； 3.运行 benchmark_selfeval_ctl.exe 程序分别统计密钥生成、封装、解封装的总耗时、平均耗时、平均 CPU 周期数和吞吐量； 5. 程序在算法运行前记录静态/基线内存占用，在多次执行过程中记录峰值内存占用,并最终生成 ctl_kem_benchmark_raw.csv， ctl_kem_benchmark_summary.csv 结果文件。
预期结果	程序能够正常运行，并输出算法的性能测试结果。包括密钥生成、封装、

	解封装操作的平均 CPU 周期数、吞吐量、静态/基线内存占用与峰值内存占用。
实测结果	程序能够正常运行，并输出性能测试结果。程序分别统计密钥生成、封装、解封装操作的平均 CPU 周期数、吞吐量静态/基线内存占用与峰值内存占用，测试次数为 100 次。 密钥生成平均 CPU 周期数为 28096354 cycles/op，吞吐量为 86 ops/s； 封装平均 CPU 周期数为 28599 cycles/op 吞吐量为 84875 ops/s； 解封装平均 CPU 周期数为 169168 cycles/op，吞吐量为 14306 ops/s。 静态/基线工作集内存平均值为 5792311 Bytes； 峰值内存平均值为 5801041.92 Byte；

测试编号	OPT-6
测试名称	CTL-3329-2048 优化实现版性能与资源消耗测试
测试目的	测试 CTL-3329-2048 优化实现版密钥生成、封装、解封装三个核心操作的平均 CPU 周期数、吞吐量、静态内存占用和峰值内存占用。
预置条件	1.CTL 优化实现版程序代码运行正常； 2.benchmark_selfeval_ctl.c 能够正常调用 CTL 算法接口； 3.测试程序可正常编译并运行。
测试步骤	1.进入 CTL-3329-2048 优化实现版程序代码目录； 2.输入 make all 生成 benchmark_selfeval_ctl.exe； 3.运行 benchmark_selfeval_ctl.exe 程序分别统计密钥生成、封装、解封装的总耗时、平均耗时、平均 CPU 周期数和吞吐量； 5. 程序在算法运行前记录静态/基线内存占用，在多次执行过程中记录峰值内存占用,并最终生成 ctl_kem_benchmark_raw.csv， ctl_kem_benchmark_summary.csv 结果文件。
预期结果	程序能够正常运行，并输出算法的性能测试结果。包括密钥生成、封装、解封装操作的平均 CPU 周期数、吞吐量、静态/基线内存占用与峰值内存占用。
实测结果	程序能够正常运行，并输出性能测试结果。程序分别统计密钥生成、封装、解封装操作的平均 CPU 周期数、吞吐量静态/基线内存占用与峰值内存占用，测试次数为 100 次。 密钥生成平均 CPU 周期数为 1567500215 cycles/op，吞吐量为 1.4ops/s； 封装平均 CPU 周期数为 147730 cycles/op 吞吐量为 16388 ops/s； 解封装平均 CPU 周期数为 33467357 cycles/op，吞吐量为 72 ops/s。 静态/基线工作集内存平均值为 7531165 Bytes； 峰值内存平均值为 7611596.80 Byte；

3.3. 传输与存储开销测试

测试编号	OPT-7
测试名称	CTL-257-512 优化实现版传输与存储开销测试
测试目的	统计 CTL-257-512 优化实现版在密钥封装过程中涉及的公钥、私钥、密文和共享密钥等数据尺寸。
预置条件	1.CTL 优化实现版程序代码运行正常； 2.算法接口能够返回或确认各类数据对象长度； 3.测试程序能够输出传输与存储开销统计结果。

测试步骤	1.进入 CTL-257-512 优化实现版程序代码目录; 2.输入 make all 生成 benchmark_selfeval_ctl.exe; 3.运行 benchmark_selfeval_ctl.exe; 4.记录公钥长度、私钥长度、密文长度和共享密钥长度;
预期结果	能够得到 CTL-257-512 优化实现版全部相关数据对象的长度, 单位为 Bytes。
实测结果	程序能够正常运行, 并输出传输与存储开销测试结果。程序分别统计公钥长, 私钥长, 密文长度, 共享密钥长度, 测试次数为 100 次。 公钥长度为 521 Bytes, 私钥长度为 2953 Bytes, 密文长度为 473 Bytes, 共享秘密值长度为 16 Bytes。

测试编号	OPT-8
测试名称	CTL-769-1024 优化实现版传输与存储开销测试
测试目的	统计 CTL-769-1024 优化实现版在密钥封装过程中涉及的公钥、私钥、密文和共享密钥等数据尺寸。
预置条件	1.CTL 优化实现版程序代码运行正常; 2.算法接口能够返回或确认各类数据对象长度; 3.测试程序能够输出传输与存储开销统计结果。
测试步骤	1.进入 CTL-769-1024 优化实现版程序代码目录; 2.输入 make all 生成 benchmark_selfeval_ctl.exe; 3.运行 benchmark_selfeval_ctl.exe; 4.记录公钥长度、私钥长度、密文长度和共享密钥长度;
预期结果	能够得到 CTL-769-1024 优化实现版全部相关数据对象的长度, 单位为 Bytes。
实测结果	程序能够正常运行, 并输出传输与存储开销测试结果。程序分别统计公钥长, 私钥长, 密文长度, 共享密钥长度, 测试次数为 100 次。 公钥长度为 1230 Bytes, 私钥长度为 6030 Bytes, 密文长度为 1006 Bytes, 共享秘密值长度为 32 Bytes。

测试编号	OPT-9
测试名称	CTL-3329-2048 优化实现版传输与存储开销测试
测试目的	统计 CTL-3329-2048 优化实现版在密钥封装过程中涉及的公钥、私钥、密文和共享密钥等数据尺寸。
预置条件	1.CTL 优化实现版程序代码运行正常; 2.算法接口能够返回或确认各类数据对象长度; 3.测试程序能够输出传输与存储开销统计结果。
测试步骤	1.进入 CTL-3329-2048 参考实现版程序代码目录; 2.输入 make all 生成 benchmark_selfeval_ctl.exe; 3.运行 benchmark_selfeval_ctl.exe; 4.记录公钥长度、私钥长度、密文长度和共享密钥长度;
预期结果	能够得到 CTL-3329-2048 优化实现版全部相关数据对象的长度, 单位为 Bytes。
实测结果	程序能够正常运行, 并输出传输与存储开销测试结果。程序分别统计公钥长, 私钥长, 密文长度, 共享密钥长度, 测试次数为 100 次。 公钥长度为 3009 Bytes, 私钥长度为 15617 Bytes,

	密文长度为 2353 Bytes，共享秘密值长度为 48 Bytes。
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